

Analysis of Algorithms

Lab 04

1. Solve the 0/1 Knapsack problem using exhaustive search. Generate the problem, capacity, weights, and values randomly for a problem size n . You could also use a benchmark dataset if you wish. Find which is the biggest problem size n that you can solve in 30s of execution. You will have to record in a table the execution time for each $n = 3, 4, 5, 6, \dots$
2. Implement a Depth First Search algorithm to print the values of each node in a graph. Decide which data structure to use: adjacency matrix or adjacency list.
3. Implement a Breadth First algorithm to print the values of each node in a graph. Decide which data structure to use: adjacency matrix or adjacency list.

Submission

Upload your solutions as a PDF file. In case of coding assignments, upload to GitHub and include the link in the PDF file. Your submission should include screenshots of the execution of your code when applicable.